FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES

CS054
(INTELLIGENT SYSTEMS)

EXERCISE
2

Sentiment Analysis

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Members:	**Name** / **Role** (4 empty rows)
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I. PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design and evaluate solutions for computing problems. [PO: e]

II. COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- To design an Intelligent System based on methods learned. [CLO: 3]

III. INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Demonstrate understanding of Sentiment Analysis.
- Design and develop a simple program about Sentiment Analysis using Python.

IV. BACKGROUND INFORMATION

Sentiment Analysis can assist us in determining the mood and feelings of the general public as well as obtaining useful information about the setting. Sentiment Analysis is the process of assessing data and categorizing it according to the research's needs.

Sentiments can be used to gain a better understanding of different events and their consequences. It goes by various names, including "sentiment analysis," "opinion mining," "opinion extraction," "sentiment mining," "subjectivity analysis," "affect analysis," "emotion analysis," and "review mining," but they all serve the same objective. We might be able to figure out what people like or want by evaluating their feelings.



TextBlob

TextBlob is a python library and offers a simple API to access its methods and perform basic NLP tasks. Textblob's sentiment function returns polarity and subjectivity as two characteristics.

The polarity ranges from [-1,1], with 1 denoting a positive statement and -1 denoting a negative one. Subjective statements usually refer to personal feelings, emotions, or judgments, whereas objective sentences refer to facts. Subjectivity is also a float with a value between 0 and 1.

V. GRADING SYSTEM / RUBRIC (please see separate sheet)

VI. LABORATORY ACTIVITY

Perform the following:

1. Add the following code:

```
from textblob import TextBlob
```

2. Declare a variable that will accept a sentence:

```
sentence = input("Enter a sentence : ")
```

3. Next initialize TextBlob

```
s = TextBlob(sentence)
```

4. Then check the polarity of the sentence by using the code below:

```
p = s.sentiment.polarity
```

5. Make a decision based on the polarity. If the polarity of the sentence is less than zero the sentiment is negative. If the polarity is equals to zero, then the sentiment is neutral. If the polarity is greater than zero, the sentiment is positive.

```
#Negative p<0, Neutral p=0, Positive p>0
if p <0:
    print("Negative")
elif p==0:
    print("Neutral")
elif p>0 and p<=1:
    print("Positive")
```

6. Make the program perform the sentiment analysis 10 times and make a summary of how many sentences are positive, neutral or negative.

OUTPUT:

```
Sentence 1
Enter a sentence: I enjoy the CSExpo event!
Polarity: 0.5
Sentiment: Positive

Sentence 2
Enter a sentence: The house is big
Polarity: 0.0
Sentiment: Neutral

Sentence 3
Enter a sentence: I hate this hot weather!
Polarity: -0.24375000000000002
Sentiment: Negative

Sentence 4
Enter a sentence: I love playing basketball
Polarity: 0.5
Sentiment: Positive

Sentence 5
Enter a sentence: It's really hot outside
Polarity: 0.125
Sentiment: Positive

Sentence 6
Enter a sentence: The food tastes so good!
Polarity: 0.875
Sentiment: Positive

Sentence 7
Enter a sentence: This bread taste so awful
Polarity: -1.0
Sentiment: Negative

Sentence 8
Enter a sentence: My cellphone is on my bag
Polarity: 0.0
Sentiment: Neutral

Sentence 9
Enter a sentence: the fuel is so expensive!
Polarity: -0.625
Sentiment: Negative

Sentence 10
Enter a sentence: I am happy that the CSExpo event is successful!
Polarity: 0.86875
Sentiment: Positive
```

```
===== SENTIMENT SUMMARY =====
Positive sentences: 5
Neutral sentences: 2
Negative sentences: 3
Total sentences: 10
```

CODE:

```
from textblob import TextBlob

# Initialize counters
positive = 0
neutral = 0
negative = 0

# Analyze 10 sentences
for i in range(10):
    print("\nSentence", i + 1)
    sentence = input("Enter a sentence: ")

    # Initialize TextBlob
    s = TextBlob(sentence)

    # Get sentiment polarity
    p = s.sentiment.polarity

    # Display polarity score
    print("Polarity:", p)

    # Determine sentiment
    if p < 0:
        print("Sentiment: Negative")
        negative += 1
    elif p == 0:
        print("Sentiment: Neutral")
        neutral += 1
    else:
        print("Sentiment: Positive")
        positive += 1

# Display summary
print("\n===== SENTIMENT SUMMARY =====")
print("Positive sentences:", positive)
print("Neutral sentences:", neutral)
print("Negative sentences:", negative)
print("Total sentences:", positive + neutral + negative)
```

CODE SCREENSHOT:

TA5_Liggayu.py > ...

```python

from textblob import TextBlob

# Initialize counters
positive = 0
neutral = 0
negative = 0

# Analyze 10 sentences
for i in range(10):
    print("\nSentence", i + 1)
    sentence = input("Enter a sentence: ")

    # Initialize TextBlob
    s = TextBlob(sentence)

    # Get sentiment polarity
    p = s.sentiment.polarity

    # Display polarity score
    print("Polarity:", p)

    # Determine sentiment
    if p < 0:
        print("Sentiment: Negative")
        negative += 1
    elif p == 0:
        print("Sentiment: Neutral")
        neutral += 1
    else:
        print("Sentiment: Positive")
        positive += 1

# Display summary
print("\n===== SENTIMENT SUMMARY =====")
print("Positive sentences:", positive)
print("Neutral sentences:", neutral)
print("Negative sentences:", negative)
print("Total sentences:", positive + neutral + negative)

```

VII. CHALLENGE

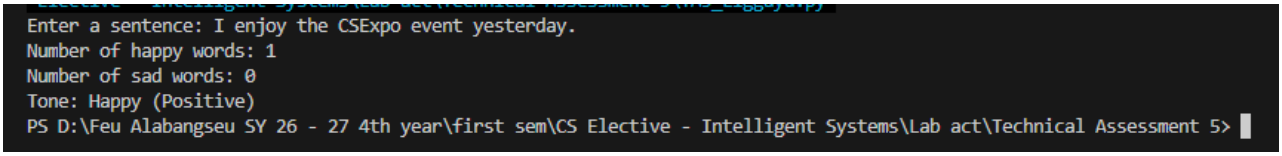
Write a program that will ask the user for a string containing a sentence. The program will then compute the happiness and sadness level of the sentence using the two functions described below. If the happiness level is higher than sadness level, then the tone of the sentence is happy. If the sadness level is higher, then the tone of the sentence is sad. Otherwise, it is neutral.

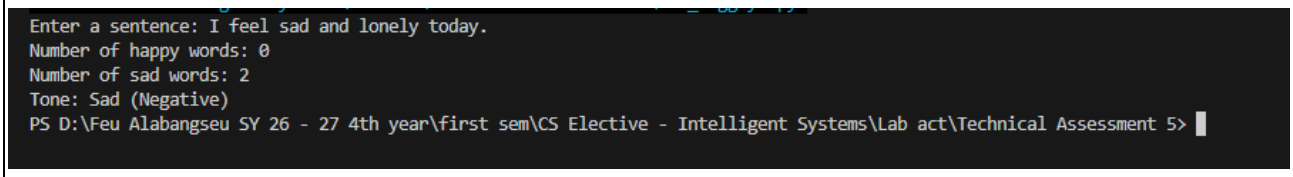
Find and print the tone of the sentence by printing the number of happy words followed by the number of sad words, followed by a simple statement of the analysis (whether positive, negative or neutral).

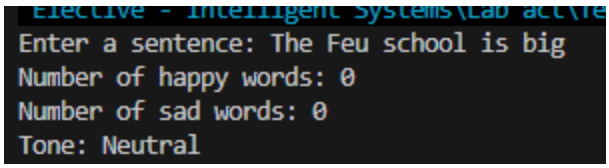
To accomplish this, you will write a function called number_happy(sentence) which returns the number of words in a given string called “sentence” that are happy. To do this, look for a public dataset containing a list of happy words. Find the total number of words in “sentence” that are in the happy words dataset.

Next, write a second function called number_sad(sentence) that works the same way but instead counts the number of sad words, also from a separate dataset.

OUTPUT:







CODE:

```
# Function to count happy words in a sentence
def number_happy(sentence):

    # List of happy words
    happy_words = [
        "happy", "joy", "love", "great", "excellent",
        "wonderful", "excited", "amazing", "glad", "fun"
    ]

    # Convert the sentence to lowercase and split it into words
    words = sentence.lower().split()

    # Initialize the happy word counter
    count = 0

    # Check each word in the sentence
    for word in words:

        # Remove punctuation marks from the word
        word = word.strip(".,!?;:")
```

```
        # Check if the word is in the happy words list
        if word in happy_words:
            count += 1

    # Return the total number of happy words
    return count


# Function to count sad words in a sentence
def number_sad(sentence):

    # List of sad words
    sad_words = [
        "sad", "unhappy", "cry", "hate", "terrible",
        "awful", "miserable", "depressed", "lonely", "bad"
    ]

    # Convert the sentence to lowercase and split it into words
    words = sentence.lower().split()

    # Initialize the sad word counter
    count = 0

    # Check each word in the sentence
    for word in words:

        # Remove punctuation marks from the word
        word = word.strip(".,!?;:")

        # Check if the word is in the sad words list
        if word in sad_words:
            count += 1

    # Return the total number of sad words
    return count


# Ask the user to enter a sentence
sentence = input("Enter a sentence: ")

# Count the happy and sad words in the sentence
happy = number_happy(sentence)
sad = number_sad(sentence)

# Display the number of happy and sad words
print("Number of happy words:", happy)
print("Number of sad words:", sad)

# Compare the number of happy and sad words
```

```
if happy > sad:

    # More happy words means a positive tone
    print("Tone: Happy (Positive)")

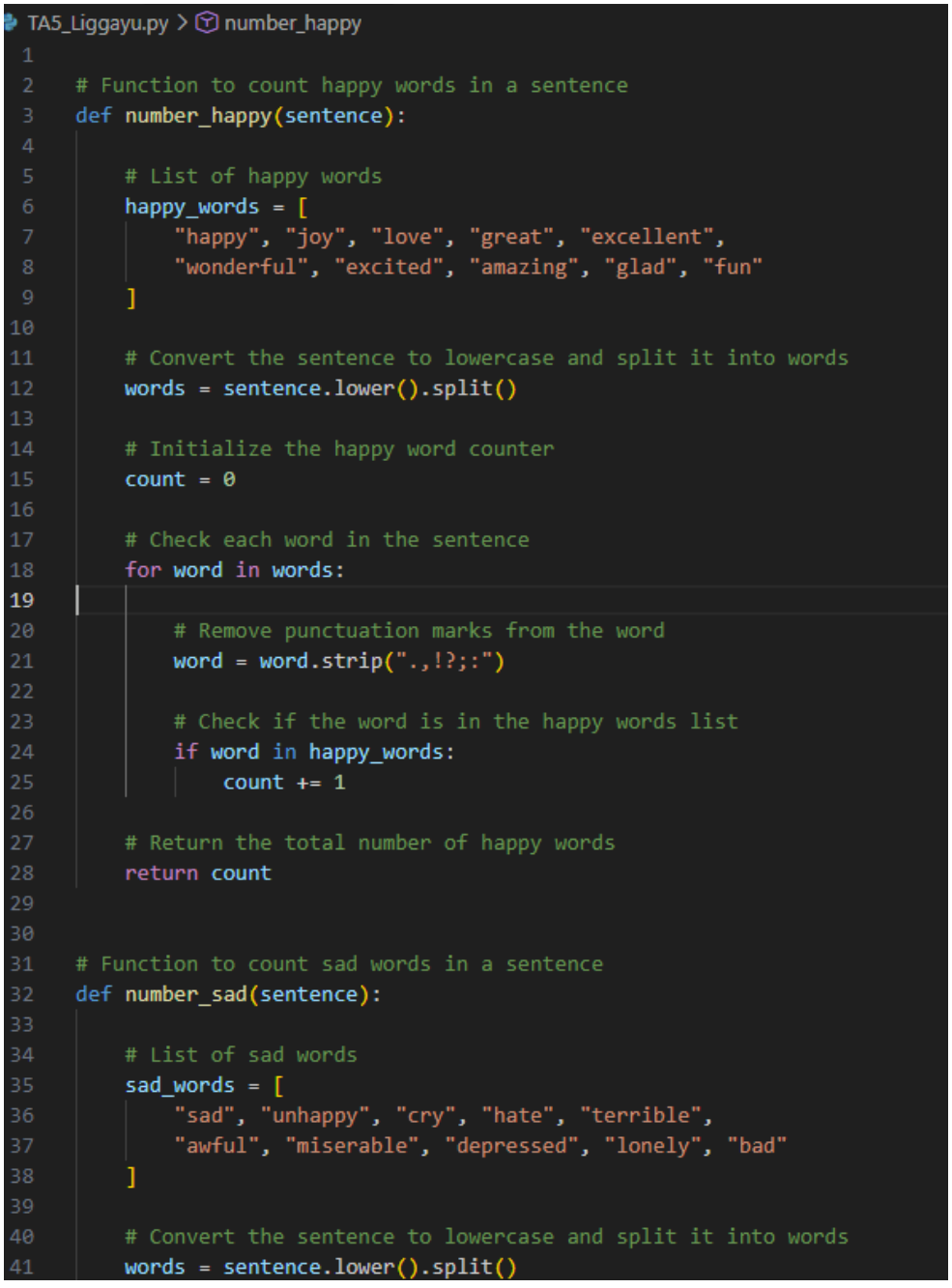
elif sad > happy:

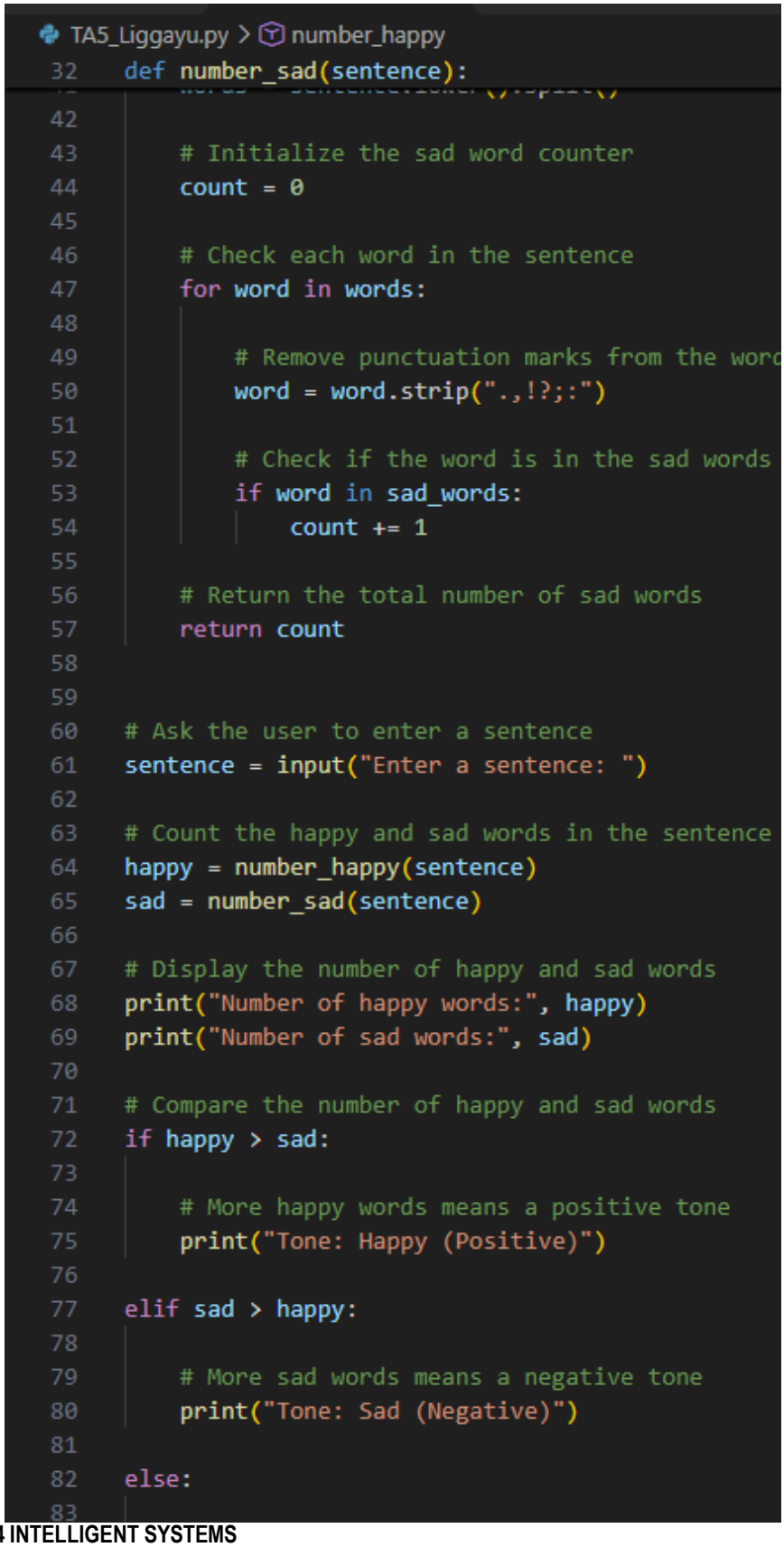
    # More sad words means a negative tone
    print("Tone: Sad (Negative)")

else:

    # Equal counts mean a neutral tone
    print("Tone: Neutral")
```

CODE SCREENSHOT:




TA5_Liggayu.py > number_happy
def number_sad(sentence):
Initialize the sad word counter
count = 0
Check each word in the sentence
for word in words:
Remove punctuation marks from the word
word = word.strip(".,!?;:")
Check if the word is in the sad words
if word in sad_words:
count += 1
Return the total number of sad words
return count
Ask the user to enter a sentence
sentence = input("Enter a sentence: ")
Count the happy and sad words in the sentence
happy = number_happy(sentence)
sad = number_sad(sentence)
Display the number of happy and sad words
print("Number of happy words:", happy)
print("Number of sad words:", sad)
Compare the number of happy and sad words
if happy > sad:
More happy words means a positive tone
print("Tone: Happy (Positive)")
elif sad > happy:
More sad words means a negative tone
print("Tone: Sad (Negative)")
else:

```
else:

    # Equal counts mean a neutral tone
    print("Tone: Neutral")

```

VIII. REFERENCES

- Artasanchez A. Joshi P.(2020). Artificial Intelligence with Python: You Complete Guide to Building Intelligent apps using Python 3.x 2nd Edition. Packt Publishing
- Attaway, Stormy (2013). MATLAB ®. A Practical Introduction to Programming and Problem Solving Third Edition. Elsevier, Inc.

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- https://research.google.com/colaboratory
- https://www.youtube.com/watch?v=RLYoEyIHL6A